(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-104 **Subject: Applied Chemistry**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Do **any six** parts: **(2x6=12)**

(a) All coking coals are caking coals but all caking coals are not coking coals.

(b) Hydrocarbons that are poor gasoline fuels are quite good diesel fuels. Explain

(c) A eutectic mixture has a definete composition and asharpmelting point, yet it is not a compound. Justify.

(d) Describe the method of preparation of the polymer, Nylon-6, 6.

(e) Why is phosphate conditioning better than the carbon aye conditioning?

(f) A water sample contains 248 mg $CaSO_4$ per liter. Calculate the hardness in terms of $CaCO_3$ equivalent.

(g) Why impure metals corrode faster than pure metal under identical conditions?

(h) A pure metal rod half immersed in water starts corroding at the bottom. Give reason.

UNIT-I

Q2 (a) A sample of coal was tested for its calorific value using Bomb's Calorimeter. Following data was obtained.
Weight of coal burnt = 0.920 gm.
Weight of water taken = 550 gm.
Water equivalent of bomb and calorimeter = 2200 gm.
Rise in temperature = 2.42 °C
Fuse wire correction = 10 Cal
Acid Correction = 50 Cal
Assuming coal to contain C = 93%, H = 6% and ash = 1%, Calculate gross and net calorific value of coal if the latent heat of condensation of steam is 580 Cal/gm. **(4)**

(b) What are the advantages of catalytic cracking over thermal cracking? **(4)**

(c) What do you understand by the term knocking in IC engine? Explain the relation between chemical structure and knocking in petrol engine. **(4)**

Q3 (a) A sample of coal was analyzed as follows. Exactly 2.5 g was weighed into a silica crucible. After heating for an hour at 110 °C, the residue weighed 2.415 g. The crucible was then covered with a vented lid and strongly heated for exactly 7 min at 950 ± 20 °C. The residue weighed 1.528 g. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to weigh 0.245 gm. Calculate the percentage of moisture, volatile matter, ash and fixed carbon in the coal sample. **(4)**

(b) What are gaseous fuels? Give some examples. What are the merits and demerits of gaseous fuels over solid and liquid fuels? **(4)**

P.T.O.